Aerospace Engineering — Graduate Programs Guide

For Olivia — a theoretical physicist from the Niels Bohr Institute with a background in string theory and a long-standing pull toward space and propulsion. This guide is a starting point: an honest map of what aerospace engineering actually contains, where the current research lives, and which programs are doing it.

You don't need to know exactly what you want to work on yet. That's part of what this guide is for.

What's here

- landscape.md — What aerospace engineering actually is: the sub-fields, what's active right now, where your physics background lands well
- programs.md — Graduate programs organized by research strength, with notes on fit for your profile
- application-guide.md — How AE graduate admissions work, and how your background translates

Your starting position

A master's in theoretical physics from NBI, with a bachelor's thesis on string theory and black holes, CERN research experience, and strong mathematical training. This is not a typical AE applicant profile — and that is not a problem.

Aerospace engineering has sub-fields that are effectively applied physics. Electric propulsion is plasma physics. Astrodynamics is celestial mechanics and optimization theory. Space environment research is space plasma physics. In these areas, your background is directly useful — not something to apologize for or translate away, but something to lead with.

The engineering fundamentals you haven't had — fluids, thermodynamics, structures — can be picked up. A strong master's program builds them in. The mathematical maturity you already have is harder to teach.

Start with landscape.md. See which sub-field pulls.